FPnAInsights: AI-Native Platform

Client brief, current-state review and Phase 1 implementation plan

Document	Brief and Phase 1 plan (Analyze with AI + AI Insights)
Date	11 September 2026
Scope	Frontend (FPNA-Frontend, React + Vite), with the backend requirements listed separately
Status	Draft for review. No application code has been changed yet.

1. Summary

The client wants FPnAInsights to be a true **AI-native** platform, with AI built into every part of the workflow instead of a single conversational "Ask AI" button, which users now expect by default. They defined **five AI features**, each for a different moment in a finance user's work, plus one **security rule** that applies to all of them.

Today, two of the five exist in basic form (Ask AI, and a simple widget-level AI popover), one partly exists (insights), and two are new (Refine with AI, Create with AI). The review also found two foundation issues that must be fixed first: the report AI ignores the filters on screen, and AI calls are sent without the user's login token, so role-based security can't be enforced.

Recommended start (Phase 1): fix the shared AI foundation, then deliver **Analyze with AI** and **AI Insights** (with report commentary and PDF export), piloted on the P&L report and then rolled out to the other reports.

2. What the client is asking

Feature	User intent	What it must do
AI Insights	"Tell me what I should know."	Proactively surface trends, variances, risks, anomalies and opportunities **for the report currently on screen**, respecting its filters, period selection and the user's permissions. Users can **copy** an insight or **add it to the report as commentary**. Added commentary becomes part of the report and is included when it is **printed or exported to PDF** for management and board reporting.
Refine with AI	"Reconsider this using what I just told you."	Applies to AI-generated **line-item variance explanations and recommendations**. The user clicks **Refine** and adds context, corrections or management judgement in plain language. The AI regenerates the text while keeping the line item's data. The user can repeat this, then **accept and save the final version as that line item's commentary**.
Ask AI	"Help me understand or solve something."	The primary conversational interface across the platform, limited to the information the user's role can see, with contextual follow-up questions. It must include **Find a Way** goal-seeking: the user states a target and constraints (e.g. *"Find a way to maintain $6M EBITDA if revenue declines 8%, without reducing headcount"*). The AI evaluates controllable drivers, proposes alternative paths, quantifies the expected outcome of each and lets the user modify constraints or

		assumptions. A chosen solution can be passed to **Create with AI** to become a scenario.
Analyze with AI	"Investigate what I'm looking at."	Available from relevant **charts, KPIs, tables and dashboard components**. It automatically inherits the current context (filters, entity, business unit, department, period, metric and comparison) without the user restating it. It finds the **root causes and quantified drivers of a variance**, with drill-down through follow-up questions and access to the supporting data.
Create with AI	"Build this for me."	The user describes a **report, analysis or scenario** in natural language. The AI builds it on screen with the right filters, dimensions, periods, metrics and comparisons. The user **reviews and customizes it before saving**. Reports can then be exported, scheduled or shared with authorized users. Scenarios become editable **Scenario Planning** objects.

> **Security rule for all five features:** existing role-based security must always apply. If a request is outside the user's permissions, the AI must not retrieve or expose the data and must reply exactly: *"You do not have access to this information as per your role."*

3. Where we are today

Feature	Status	What exists	Gap to the client's requirement
Ask AI	**Mostly exists**	Full-page assistant (`AIChatBot.jsx`, `/api/v1/ai/assistant/chat`) with saved chat history and server-side topic routing. Report-level Ask AI drawer on 13 reports and a small chatbot (`/api/v1/ai`).	No Find a Way goal-seeking. No role-based refusal message. No hand-off to a scenario.
Analyze with AI	**Partial**	"Ask AI" popover on KPI tiles and chart cards (`WidgetAskAI.jsx`, `chartAiContext.js`) with preset questions only.	Sends only the plotted numbers, with no filters, entity, business unit, department or comparison. No free-text follow-up, no root-cause analysis or quantified drivers, no supporting data view. Not available on table rows.
AI Insights	**Partial**	Separate "AI-driven Insights & Alerts" pages, an alert-insights API, and locally generated insight text on a few reports.	No insights panel on each report. No copy / add-to-report commentary. Financial reports have no PDF export to include commentary in.
Refine with AI	**New**	Nothing equivalent.	Needs AI line-item variance explanations, the refine loop and saved line-item commentary.

Create with AI	**New**	The targets exist but are manual: Custom Report Builder and Scenario Modeling.	Needs natural-language to report/scenario generation, review-before-save, and saving as editable objects.

4. Key findings from the code review

1. **The report AI ignores what's on screen.** The report Ask AI drawer receives only a report id and **re-fetches the report with fixed defaults** (all periods, default sort). Filters the user applied (entity, cost/profit centre, customer, region, product, version, period, comparison) are ignored, so answers can describe different numbers from the ones on screen. This contradicts the client's first requirement.
2. **AI requests carry no login token.** Calls to `/api/v1/ai` are sent without an `Authorization` header, so the backend can't know who is asking or enforce role-based access.
3. **The widget AI is a preset-question popover** with no filter context, no free-text input and no root-cause output.
4. **The main financial reports have no PDF export.** jsPDF is installed and used only on the Smart Alerts and Supply Chain pages, so "include insights in the PDF" has nothing to attach to yet.

> **Implication:** items 1 and 2 are prerequisites for every AI feature in the client's table. Phase 1 fixes them first, in a shared layer the later phases reuse.

5. Phase 1 plan: Analyze with AI + AI Insights

5.0 Assumptions (to confirm)

- **Pilot on the P&L report** end to end, then roll the same pattern out to the other 12 reports.
- **Insights are generated when the panel is opened** and cached per filter set, not on every page load, to control AI cost.
- **Commentary is stored in the browser (localStorage)** behind a small storage module until the backend commentary API exists. Switching later changes one file.

5.1 Shared AI foundation (new folder `src/ai/`)

- `aiClient.js`: one function for all AI calls. It always sends the login token, plus the feature name, report id, applied filters and conversation history. A "no access" response from the backend (HTTP 403) is shown as the exact client message: *"You do not have access to this information as per your role."* The existing `askAi.js` helper becomes a thin wrapper, so current callers keep working.
- `reportContext.js`: the report-context builders move here from the Ask AI drawer, plus a `describeFilters` helper that turns applied filters into a readable line (e.g. "Entity: X · Period: Q2 · Compared to budget").
- `useReportAIContext.js`: a hook each report page calls with the **data already loaded on screen and the applied filters**. It returns the AI context and a cache key (report, year, currency and filters).

5.2 Ask AI uses on-screen data

- The Ask AI drawer accepts the report's on-screen **data** and **filters**. When they are provided, the AI answers from exactly what the user sees. Without them the old behaviour stays, so all reports keep working during rollout.
- The conversation resets when filters change (not only year/currency).
- A free-text question box is added under the preset questions, with follow-up history.
- **Pilot:** the P&L page passes its loaded rows and applied filters.

5.3 AI Insights panel and report commentary

- **AI Insights panel** (collapsible "✦ AI Insights" card on the report): when opened, it generates 3–6 insights from the on-screen report, each tagged *trend / variance / risk / anomaly / opportunity*. Each insight has **Copy** and **Add to report**. It includes Refresh, loading/empty/error states and the access-denied message.
- **Commentary section** on the report: shows the added insights. Items can be edited, removed and reordered, and each shows its source (AI or manual), author and date. It's saved per report, period and filter set.

5.4 Analyze with AI

- The widget AI popover becomes **Analyze with AI** (the existing component name stays as an alias so the 4 dashboards keep working).
- On open, it **automatically runs a root-cause analysis**: a short summary, the **quantified drivers** (driver, impact, share of the variance) and a collapsible **"Data used"** table showing the exact numbers analysed. Free-text follow-up questions keep the context.
- Dashboards (Financial Overview charts and KPI tiles, Key Financial KPIs, Budget Overview) pass their full context: year, currency, entity and filters, not just the plotted series.
- **Table rows:** each P&L line item gets an Analyze icon next to the existing drill-down. It sends the line's actual, comparison, variance, variance % and monthly values.

5.5 PDF export with commentary

- A shared **Export PDF** function (jsPDF + autoTable, same approach as the existing Smart Alerts export): report title, applied filters, the report table, then the **Commentary** section.
- An **Export PDF** button on the P&L toolbar next to the existing CSV export, and print styles so the browser's print output also includes the commentary.

5.6 Rollout after the pilot

Apply the same wiring (on-screen data for Ask AI, AI Insights, Commentary, Export PDF, row-level Analyze) to the other report pages: Revenue, Customer Profitability, Cash Flow, Balance Sheet, Operating Expenses, Project Costing, Fixed Assets, Sales Funnel and Payroll Reconciliation.

6. Backend requirements

The backend is a separate codebase. Phase 1 works with the existing `/api/v1/ai` endpoint, but these changes are needed for security and for commentary shared across users and exports:

Endpoint	Change / purpose	Needed for
`POST /api/v1/ai`	Accept the login token, feature, report id, filters and history. Enforce the user's role and data scope. Return **403** `{code: "access_denied"}` for anything outside the role.	All features (security rule)
`POST /api/v1/ai/insights` (optional)	Return structured insights `{type, title, text}` for a report and filter set. Phase 1 can work without it by prompting the existing endpoint for JSON.	AI Insights
`/api/v1/company/financial/report-commentary` (GET / POST / PATCH / DELETE)	Store report and line-item commentary per report, year, currency and filter set. Replaces browser storage, shares commentary across users, and lets scheduled report exports include it.	AI Insights, Refine with AI

7. Files

Type	Files
New	`src/ai/aiClient.js`, `src/ai/reportContext.js`, `src/ai/useReportAIContext.js`, `src/ai/commentaryStore.js`, `src/ai/exportReportPdf.js`, `src/components/AI/ReportInsights.jsx`, `src/components/AI/ReportCommentary.jsx`, `src/components/AI/AnalyzeWithAI.jsx`
Modified	`src/pages/FinancialReports/AskAI.jsx`, `src/components/Dashboard/WidgetAskAI.jsx` (becomes an alias), `src/utils/askAi.js`, `src/pages/FinancialReports/pnl/PnLPage.jsx`, the four dashboard callers

	(`FinancialCharts.jsx`, `FinancialMetrics.jsx`, `KeyFinancialKPIs.jsx`, `BudgetOverview.jsx`), then the rollout report pages.

8. Testing and acceptance

- **Automated:** lint and build checks, plus unit tests (Vitest, already set up) for the filter description, the report context (includes the applied filters, respects the size limit), commentary storage and the access-denied message mapping.
- **Acceptance on the P&L as a logged-in user:**
  1. Apply entity and period filters, then ask AI. Answers reference the filtered figures.
  2. Open AI Insights, copy one insight and add another to the report. The commentary appears and survives a reload.
  3. Export PDF. The file contains the table and the commentary.
  4. Analyze a variance row. Drivers, data used and a follow-up question all work.
  5. Simulate a no-access response. The exact client message is shown.
- **Regression:** the other 12 report pages keep their current Ask AI behaviour until they are rolled out.

9. Roadmap: phases 2–5 (for discussion)

Phase	Scope	Main dependency
1	Foundation, Analyze with AI, AI Insights, commentary, PDF export	Token + 403 handling on `/api/v1/ai` (can start before it lands)
2	Refine with AI on line-item variance explanations	Backend commentary API (line-item level)
3	Ask AI: Find a Way goal-seeking	Backend driver model / solver over controllable drivers
4	Create with AI: reports	Natural-language to Custom Report Builder specification; save, schedule and share
5	Create with AI: scenarios (including hand-off from Find a Way)	Scenario Modeling API to create editable scenarios

10. Open decisions

- Confirm the pilot report (P&L proposed) and the rollout order.
- Commentary: start with browser storage, or wait for the backend commentary API so it's shared across users from day one?

- Insights: generate when the panel is opened (proposed, lower cost) or automatically on page load?
- Backend timeline for the token/role enforcement on `/api/v1/ai`, which the security rule depends on.